# HOMA Metabolic Assessment Calculator

The Dr. Nehru Method — India-Specific Cutoffs — HOMA Healthcare Center, Hyderabad

## Patient Details

**Full Name**: subbaraosir indu sept26

**Age**: 56

**Sex**: Male

**Registration No.**: 000120260610

**Date**: 08-10-2026

## Anthropometry

**Height (cm)**: 168

**Weight (kg)**: 86

**Waist Circumference (cm)**: 112

**Hip Circumference (cm)**: 125

**Neck Circumference (cm)**: 43

**Calf Circumference (cm)**: 32

**Mid-Arm Circumference / MUAC (cm)**: 39

## Bedside Sarcopenia Tests

**Handgrip Strength, dominant hand (kg)**

**4-metre Gait Time (seconds)**

**Chair Stand Test — 5 rises (seconds)**

32

Normal cutoff: Men ≥27kg, Women ≥16kg (AWGS 2019)

14

Speed below 1.0 m/s (i.e. time above 4 sec) — slow, sarcopenia risk (AWGS 2019)

18

Above 12 seconds — impaired

## Family History

- [x] BP (Hypertension)
- [ ] Sugar (Diabetes)
- [ ] Heart Disease
- [ ] Cancer
- [ ] Brain Stroke
- [ ] Menopause (for female patients)

## Core Metabolic Markers

**Fasting Glucose (mg/dL)**

89

**Fasting Insulin (µIU/mL)**

33.4

Typical range <25 µIU/mL — flag if far outside this

**Fasting Triglycerides (mg/dL)**

383

**HbA1c (%)**

5.6

**LDL Cholesterol (mg/dL)**

174

**HDL Cholesterol (mg/dL)**

41

**C-Peptide (ng/mL)**

2.8

**Pro-insulin (pmol/L)**

9

Normal below 11 pmol/L — above signals beta-cell stress

**Faecal Elastase (mcg/g) — if malabsorption suspected**

0

>200 normal · 100-200 moderate insufficiency · <100 severe insufficiency

## Micronutrients

**Vitamin B12 (pg/mL)**

**Folic Acid (ng/mL)**

**25(OH) Vitamin D (ng/mL)**

514 | 12 | 35

**Serum Calcium (mg/dL)** 9

**Serum Ferritin (ng/mL)** 236

**Transferrin Saturation (%)** 40

## Lifestyle

**Cigarettes per day** 0

**Years smoked** 0

**Alcohol volume per drink (ml)** 30

**Alcohol % ABV (Beer~5, Whisky~40, Wine~12, Toddy~6)** 40

**Drinks per week** 12

**Usual Bedtime** 10:00 PM – 12:00 Midnight

**Average Daily Steps** 7,500 – 12,000 (Moderate)

## Comorbidities

- [ ] Chronic Kidney Disease
- [ ] Fatty Liver
- [ ] Angiography / PTCA / Stent
- [ ] Brain Stroke

## Medications & Duration of Use

**Statins**

**BP Tablets**

**Sugar (Diabetes) Tablets**

Less than 12 months | 5 – 12 years | 1 – 5 years

## Body Composition (BIA / DEXA / Imaging — fill if available)

**SMI (kg/m²)**

**Visceral Fat Area — BIA/MRI/CT (cm²)**

**Phase Angle (degrees)**

**Body Fat (%)**

Save to Records

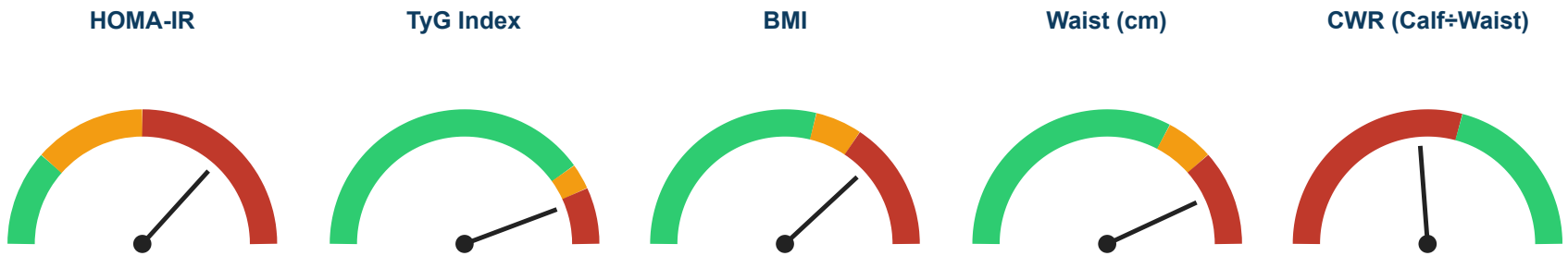
## Calculated Results

| Measure | Value | Flag | Note |
|---|---|---|---|
| HOMA-IR | 7.34 | High | India cutoff: above 2.3 |
| TyG Index | 9.74 | High | India cutoff: above 8.82 (ln scale) |
| BMI | 30.5 kg/m² | High | India screening cutoff: above 23 |
| Waist-to-Hip Ratio | 0.90 | Normal | |

| | | | |
|---|---|---|---|
| Waist-to-Height Ratio | 0.67 | High | |
| Waist Circumference | 112 cm | High | |
| CWR (Calf ÷ Waist) | 0.29 | High | Below 0.35 — high sarcopenia risk (Dr. Nehru Method) |
| LDL Cholesterol | 174 mg/dL | High | |
| HDL Cholesterol | 41 mg/dL | Normal | |
| Neck Circumference | 43 cm | Normal | |
| Handgrip Strength | 32 kg | Normal | Men <27.5kg (Indian Delphi 2025), Women <18kg (AWGS 2019) |
| 4-metre Gait Speed | 0.29 m/s | High | Below 1.0 m/s — slow, sarcopenia risk |
| Chair Stand Test (5 rises) | 18 sec | High | Above 12 sec — impaired |
| Pro-insulin | 9 pmol/L | Normal | Normal below 11 pmol/L |
| Pack-Years | — | | |
| Alcohol Units / Week | 11.4 | Normal | |

| | | | |
|---|---|---|---|
| Vitamin B12 | 514 pg/mL | Normal | Deficient <200, Borderline 200-300 pg/mL |
| Vitamin D (25-OH) | 35 ng/mL | Normal | |
| Ferritin | 236 ng/mL | Normal | |
| Sleep Timing | 10 PM–12 Midnight | Borderline | |
| Daily Steps | 7,500–12,000 (Moderate) | Normal | |
| Comorbidities | None reported | None | |
| Family History | BP | Present | |
| Statin Use | <12 months | | |
| BP Medication | 5–12 years | | |
| Diabetes Medication | 1–5 years | | |

## Speedometer Gauges




| HOMA-IR | TyG Index | BMI | Waist (cm) | CWR (Calf÷Waist) |
|---|---|---|---|---|
| 7.34 | 9.74 | 30.5 | 112 | 0.29 |
| Green ≤2.3 · Red >5 | Green ≤8.82 · Red >9.5 | Green ≤23 · Red >27.5 | Green ≤85 · Red >100 | Red <0.35 (high risk) |

All India-specific cutoffs (HOMA-IR, TyG Index, sarcopenia thresholds) per Dr. Muddu Surendra Nehru MD, Professor of Medicine, HOMA Healthcare Center, RSSDI Life Member 8058. Phenotype (TOFI/NWCO/MONW) thresholds are proposed operational cutoffs from published Indian cohort data,

not universal consensus. This tool supports clinical judgement — it does not replace it.

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